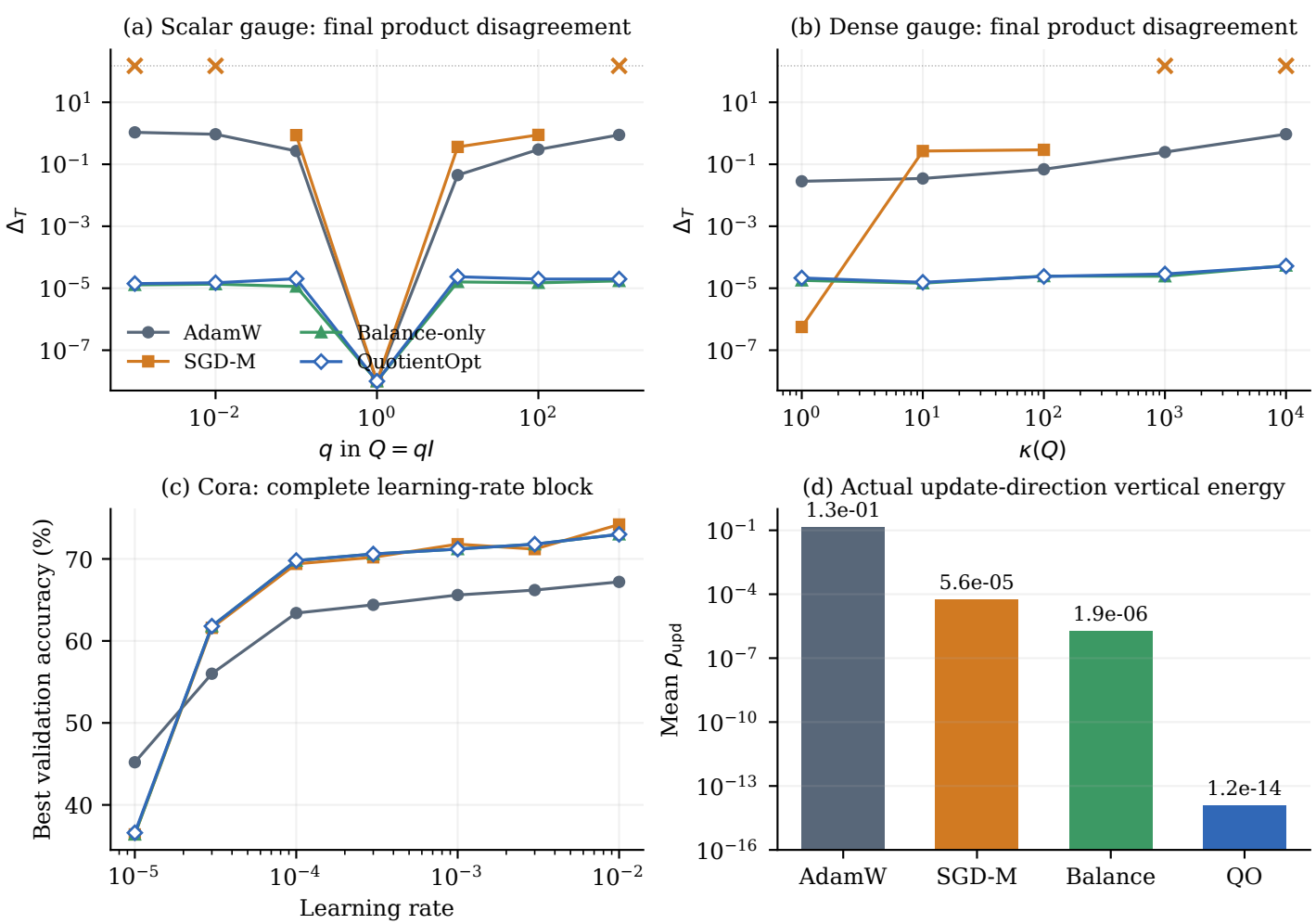
(a) Scalar gauge: final product disagreement
Δ_T
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
AdamW
SGD-M
Balance-only
QuotientOpt
10^{-2}
10^{0}
10^{2}
q in $Q = qI$
(b) Dense gauge: final product disagreement
Δ_T
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
$\kappa(Q)$
(c) Cora: complete learning-rate block
Best validation accuracy (%)
70
60
50
40
10^{-5}
10^{-4}
10^{-3}
10^{-2}
Learning rate
(d) Actual update-direction vertical energy
Mean ρ_{upd}
10^{-1}
10^{-4}
10^{-7}
10^{-10}
10^{-13}
10^{-16}
1.3e-01
5.6e-05
1.9e-06
1.2e-14
AdamW
SGD-M
Balance
QO